VdistCox: Vertically distributed Cox proportional hazards model with hyperparameter optimization

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Introduction

- The most common model in survival analysis is the Cox proportional hazards model (Cox model).
- The various features required for predicting a patient's prognosis have mutually exclusive characteristics in the form of vertically partitioned data. A patient's prognosis can be predicted more precisely by using information about the same patient from different institutions.
- In this study, we developed a vertically distributed Cox model (referred as to VdistCox) with neural network structure while avoiding the transmission of patient features, including hyperparameter optimization in a one-shot manner.

Method

- Cox model [1] in non-partitioned data

$$h_i(t) = h_0(t)exp(f(x_i))$$

where $f(\boldsymbol{x_i})$ can be any function of $\boldsymbol{x_i}$, and for a standard Cox model, $f(x_i) = x_i\beta$.

- We considered the Cox model with neural networks by replacing $f(x_i)$ with the Extreme Learning Machine (ELM) [2] output, as a distributed cox model.
- The proposed model is communication efficient without iterative communication between the server and sites.

Notation	Description
K	Number of sites
n	Number of patients for model training
$\tilde{n}$	Number of patients for model validation
N $(= n + \tilde{n})$	Number of patients
X	$(n \times M)$ Feature matrix for model training
$\tilde{X}$	$(\tilde{n} \times M)$ Feature matrix for model validation
M	Number of features distributed across K sites
L	Number of nodes
S	Number of randomly generated input weight
$R(s)$	$((M+1) \times L)$ Random matrix of s-th input weight
$\beta(s)$	Output weight of s-th random input weight. L- dimensional vector
$g(.)$	Activation function
M_k	Number of features for the k-th party
$R_k(s)$	$(M_k \times L)$ Random matrix of s-th input weight at k site, k = 2, …, K
$(X^k)_{n\times M_k}$	$(n \times M_k)$ Feature matrix of k party for model training, k = 2, …, K
$(\tilde{X}^k)_{\tilde{n}\times M_k}$	$(\tilde{n} \times M_k)$ Feature matrix of k party for model validation, k = 2, …, K

- $T_k(s) = X^{\boldsymbol{k}} R_k(s)$, $\tilde{T}_k(s) = \tilde{X}^{\boldsymbol{k}} R_k(s)$ • H(s)=g(T(s)) , H(s)=g(T(s))
- The negative log-partial likelihood of VdistCox

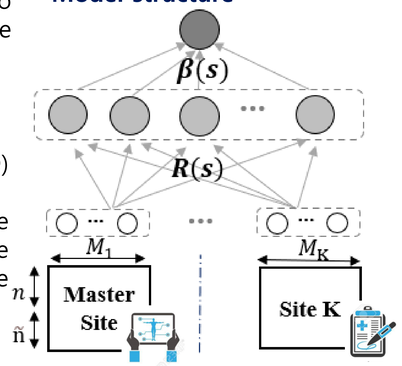
$$L(\boldsymbol{\beta}(\boldsymbol{s})) = \sum_{t=1}^{T} d_t \log\left(\sum_{j\in\mathcal{R}_t} \exp\left(\sum_{l=1}^{L} \beta_l(s)\, g(x_j, b_l(s), r_l(s))\right)\right) - \sum_{t=1}^{T}\sum_{u\in\mathcal{D}_t}\left(\sum_{l=1}^{L} \beta_l(s)\, g(x_u, b_l(s), r_l(s))\right) + \lambda\|\boldsymbol{\beta}(\boldsymbol{s})\|_2^2$$

- There are three hyperparameters: $g(.)$, λ, and L
- The activation function and the regularization parameter can be adjusted at the master site.

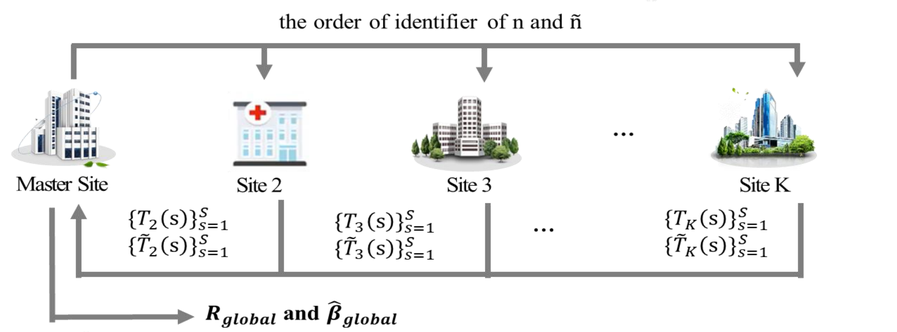
- Master site plays the role of a server to aggregate the intermediate results from the sites and derive the final model.
- VdistCox requires the following assumptions before implementation:

1. There is a unique identifier (e.g., study ID) shared across institutions.
2. It is not necessary to store event and time outcome information in every site. One of the sites stored the outcome should be the master party.

- **Model structure**



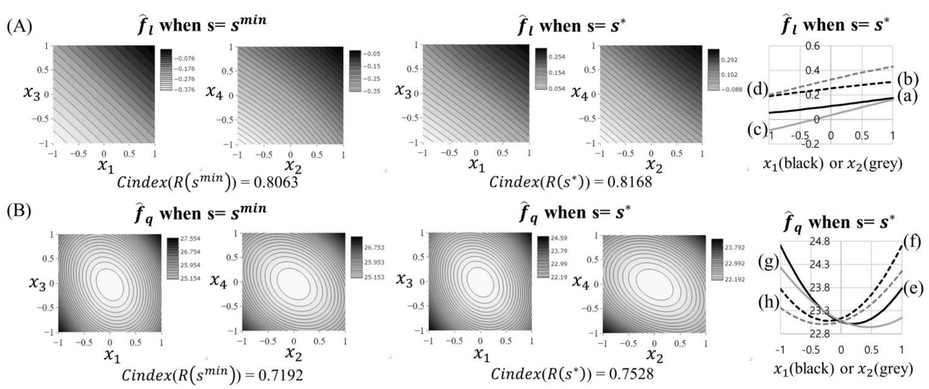
- **Process of communication.**



- $\{R_k(s)\}_{s=1}^{S}$ of size $(M_k \times L_{max})$ is generated at each site by setting the maximum number of nodes, L_{max}. $R_k(s)$ is divided into various sizes of $(M_k \times L_1)$, $(M_k \times L_2)$, …, and $(M_k \times L_{max})$.

Results

- **Simulated Data**
- f_l (Linear): $\beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$
- f_q (Quadratic + interaction): $\beta_1 {x_1}^2 + \beta_2 {x_2}^2 + \beta_3 {x_3}^2 + \beta_4 {x_4}^2 + \beta_5 x_1 x_3 + \beta_6 x_2 x_4$
- f_l : $[\beta_1, \beta_2, \beta_3, \beta_4]$= [0.5, 1, 0.5, 1]
- f_q: $[\beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6]$= [2, 1, 2, 1, 1, 1]



- Simulation results under (A) f_l and (B) f_q. Site 1 stores x_1 and x_2 and Site 2 stores x_3 and x_4. S=100, $s^* = argmax_s Cindex(R(s))$, $s^{min} = argmin_s Cindex(R(s))$, The true functions of a (black solid), b (black dashed), c (grey solid), d (grey dashed), e (black solid), f (black dashed), g (grey solid), and h (grey dashed) are $f(x) = 0.5x_1 - 0.5$, $f(x) = 0.5x_1 + 0.5$, $f(x) = x_2 - 1$, $f(x) = x_2 + 1$, $f(x) = 2{x_1}^2 - x_1 + 2$, $f(x) = 2{x_1}^2 + x_1 + 2$, $f(x) = {x_2}^2 - x_2 + 1$, and $f(x) = {x_2}^2 + x_2 + 1$, respectively.

- **Real Data:** Electronic Intensive Care Unit (eICU) Collaborative Research Database.
- 19 hospitals were included in 2,486 stays. We randomly selected 486 stays as the test set, and divided 2,000 stays 8:2 into the training and validation sets.
- Two vertical sites were assumed. Site 1 was a master site with 14 features and outcomes, where site 2 was a site with only 13 features.
- Hyperparameter settings of 5 hidden nodes × 5 activation functions × 6 regularization parameters= 150 were explored.
- S = 100, $s^* = argmax_s Cindex(R(s))$, $s^{min} = argmin_s Cindex(R(s))$, and s^{med} is s when $Cindex(R(s))$ has a median value.

VdistCox	ELU				Sigmoid			
	300/1000 (L/λ)		500/0.1 (L/λ)		500/10 (L/λ)		500/0.1 (L/λ)	
	validation	test	validation	test	validation	test	validation	test
s^{min}	0.8170	0.7149	0.4216	0.4458	0.7938	0.7162	0.2563	0.4253
s^{med}	0.8296	0.7204	0.5419	0.5086	0.8144	0.7154	0.3686	0.4686
s^*	**0.8466**	**0.7294**	0.7440	0.6017	0.8422	0.7159	0.7539	0.6502
Standard Cox model	test: 0.7160							

- The performance in the test set was the highest at ELU/L = 300/λ = 1000, which had the highest performance in the validation set; s^{med} and s* in this setting showed higher performance than the standard Cox model.

Conclusion

- The model proposed in this study, VdistCox, is communication-efficient vertically distributed Cox model by sharing once the intermediate results
- We proposed generating random input weights multiple times and a hyperparameter tuning process in a one-shot mannner.
- Considering multiple random input weights and hyperparameter optimization can be an effective means for a stable vertically distributed Cox model.

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References

[1] K. Dietz et al., Stat. Biol. Health Logistic Regression SelfLearn. Text., vol. 2, pp. 102–124, 2002.

[2] Huang, G. B., Zhu, Q. Y., & Siew, C. K. (2004, July). Extreme learning machine: a new learning scheme of feedforward neural networks. In 2004 IEEE international joint conference on neural networks (IEEE Cat. No. 04CH37541) (Vol. 2, pp. 985-990). Ieee.